LILA F. D. PERKINS

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April 23, 2026

EDUCATION

University of Washington — Sep 2025 – Present
Ph.D. Student in Applied Mathematics — *Seattle, WA*
Advisor: Baosen Zhang (Department of Electrical & Computer Engineering)

University of Washington — Jun 2026 (Expected)
Master of Science in Applied and Computational Mathematics — *Seattle, WA*
GPA: 3.91/4.00

University of Michigan — May 2025
Bachelor of Science in Honors Mathematics with High Distinction — *Ann Arbor, MI*
Minor in Complex Systems
College of Literature, Science, and the Arts (LSA) Residential College and Honors Program
GPA: 3.98/4.00

Budapest Semesters in Mathematics (BSM) — Fall 2023
Completed with Honors Recognition — *Budapest, Hungary*
Competitive semester program emphasizing the Hungarian method of mathematics instruction.
Coursework: Advanced Combinatorics, Theory of Computing, Non-Euclidean Geometries.
GPA: 4.0/4.0

RESEARCH EXPERIENCE

University of Washington — Sep 2025 – Present
Graduate Student Researcher, Dept. of Applied Mathematics — *Seattle, WA*
Advisor: Prof. Baosen Zhang (Dept. of Electrical and Computer Engineering)

- Budget-Constrained Resource Allocation: Proved the existence and uniqueness of competitive equilibrium in electricity markets with budget constrained users; demonstrated equivalence to a convex welfare maximization problem with modified utility functions (work submitted to *IEEE Control Systems Letters* and *IEEE CDC 2026*).
- Budget Constraints in Networked Markets: Currently extending frameworks to model strategic data centers participating in both compute and electricity markets across multiple geographic regions; analyzing conditions for equilibrium existence and efficiency loss in Cournot settings.

Park City Mathematics Institute/Institute for Advanced Study — July 2025
Undergraduate Summer School — *Park City, UT*

- Conducted research exploring eigenpolytopes as a bridge between abstract algebra and graph theory within a three-week program on Extremal and Probabilistic Combinatorics.

National Renewable Energy Laboratory — Summer 2024
DOE Science Undergraduate Laboratory Internship — *Golden, CO*
Advisor: Dr. Bernard Knueven

- Grid Resilience: Developed a computationally efficient numerical linear algebra approach using the Woodbury Matrix Identity to identify potential new transmission lines that improve grid resilience with respect to contingency analyses.
- Implementation: Implemented solutions in Julia and tested on the IEEE Power Grid Library; presented at the NREL Intern Symposium and the University of Michigan SIAM Student Mini-Symposium.

University of Michigan — Winter 2024
MATH 389 Explorations in Mathematics — *Ann Arbor, MI*
Advisors: Prof. Aaron Pixton and Dr. Patrick Boland

- Analyzed strategies for the Taxman number factoring game using number theory, graph theory, and computational methods.

- Explored patterns arising from agent motion on varying graph structures using algebraic and computational methods.

National Renewable Energy Laboratory Summer 2023
DOE Science Undergraduate Laboratory Internship *Golden, CO*
Advisor: Dr. Gayathri Krishnamoorthy

- Resilience Planing: Developed a grid resilience database and a FEMA Expected Annual Loss calculator for utility service territories to inform infrastructure investment strategies.
- Dissemination: Co-authored three DOE Technical Reports on distribution utility resilience planning for wildfires, hurricanes, and winter storms.

PUBLICATIONS

Perkins, L. and Zhang, B. "Resource Allocation in Electricity Markets with Budget Constrained Customers." *Submitted to IEEE Control Systems Letters and IEEE Conference on Decision and Control*, 2026. arXiv:2603.20277

Keen, J., Matsuda-Dunn, R., Krishnamoorthy, G., Clapper, H., **Perkins, L.**, Leddy, L., and Grue, N. "Current Practices in Distribution Utility Resilience Planning for Wildfires." DOE Technical Report, Aug. 2024. DOI:10.2172/2478838

Keen, J., Matsuda-Dunn, R., Krishnamoorthy, G., Clapper, H., **Perkins, L.**, Leddy, L., and Grue, N. "Current Practices in Distribution Utility Resilience Planning for Hurricanes and Non-Winter Storms." DOE Technical Report, Aug. 2024. DOI:10.2172/2478839

Keen, J., Matsuda-Dunn, R., Krishnamoorthy, G., Clapper, H., **Perkins, L.**, Leddy, L., and Grue, N. "Current Practices in Distribution Utility Resilience Planning for Winter Storms." DOE Technical Report, Aug. 2024. DOI:10.2172/2478837

PRESENTATIONS

Perkins, L. and Knueven, B. "A mathematical approach for identifying new transmission lines to enhance grid resilience." University of Michigan SIAM Student Mini-Symposium, Ann Arbor, MI. Sep 2024. *[Poster]*

Perkins, L. and Knueven, B. "A mathematical approach for identifying new transmission lines to enhance grid resilience." NREL Intern Symposium, Golden, CO. Aug 2024. *[Poster]*

Perkins, L. and Krishnamoorthy, G. "Grid resilience investment database: supplement to Distribution resilience framework for hazard cases." NREL Intern Symposium, Golden, CO. Aug 2023. *[Poster]*

AWARDS AND HONORS

DOE Computational Science Graduate Fellowship Apr 2026
Awarded for proposed research on optimization methods for scheduling compute workloads across heterogeneous, geographically distributed infrastructure.

NSF Graduate Research Fellowship Apr 2026
Awarded for proposed research on optimization methods for AI workloads and electricity market integration.

Boeing Graduate Fellowship 2025 – 2027
University of Washington, Department of Applied Mathematics.

Phi Beta Kappa Winter 2025
University of Michigan.

Mathematics Merit Scholarship 2024 – 2025
University of Michigan, Department of Mathematics.

Phi Beta Phi Winter 2023
University of Michigan.

James B. Angell Scholar 2023 – 2025
University of Michigan. Awarded to students earning all A or A+ grades in a full-credit semester.

J. Perry Bartlett STEM Scholar 2021 – 2025
Four-year college scholarship to school of choice.

TEACHING AND EMPLOYMENT

University of Michigan, Department of Mathematics — Winter 2025
Course Assistant — MATH 389: Explorations in Mathematics — *Ann Arbor, MI*
Mentored groups of undergraduates on mathematics research projects.

University of Michigan, Department of Mathematics — Winter 2025
Grader — MATH 404: Intermediate Differential Equations — *Ann Arbor, MI*
Graded weekly homework for undergraduate differential equations course.

University of Michigan, Department of Mathematics — 2022 – 2023
Peer Tutor — MATH 217: Linear Algebra — *Ann Arbor, MI*
Provided one-on-one tutoring for undergraduate linear algebra students.

BrightLine Group — Jul – Oct 2022
Data Analysis Intern — *Boulder, CO*
Analyzed residential solar installation efficiency and evaluated California's Single-Family Affordable Solar Homes (SASH) and Disadvantaged Communities – Single-Family Solar Homes (DAC-SASH) programs effectiveness. Drafted methodology section of client report.

SERVICE AND OUTREACH

KiloWatts for Humanity — Oct 2025 – Present
Volunteer — *Seattle, WA*
Volunteer with a Seattle-based organization providing renewable energy access to communities without reliable electricity.

Outdoors for All — Winter 2025
Volunteer Adaptive Ski Instructor — *Seattle, WA*
Teaching adaptive skiing to participants with physical and/or cognitive disabilities through a Seattle-based nonprofit making outdoor recreation accessible.

Special Olympics Michigan, Area 20 — 2022 – 2025
Volunteer — *Ann Arbor, MI*
Assisted with weekly basketball practices for athletes with intellectual disabilities.

Ignite Adaptive Sports — 2019 – 2021
Volunteer Junior Instructor — *Boulder, CO*
Taught adaptive skiing to participants with physical and/or cognitive disabilities.

SELECTED ADVANCED COURSEWORK

University of Washington
Optimization: Fundamentals and Applications; Networks and Combinatorial Optimization; Numerical Linear Algebra; Numerical Analysis; Numerical Methods for Partial Differential Equations

University of Michigan (all graduate-level)
Computer Modeling of Complex Systems; Theory of Complex Systems; Numerical Methods for Differential Equations; Probability Theory; Sustainable Energy Systems; Advanced Energy Solutions

TECHNICAL SKILLS AND LANGUAGES

Programming: Julia, Python, MATLAB, C++, Mathematica, LaTeX, Excel, Igor Pro, SOLIDWORKS, AutoCAD, NLR System Advisor Model

Languages: English (native); Spanish (fluent); Portuguese (proficient); Hungarian (basic)

PROFESSIONAL MEMBERSHIPS

American Mathematical Society (AMS) — 2026 – Present
Institute of Electrical and Electronics Engineers (IEEE), Graduate Student Member — 2026 – Present
IEEE Power & Energy Society (PES)
Society for Industrial and Applied Mathematics (SIAM) — 2026 – Present